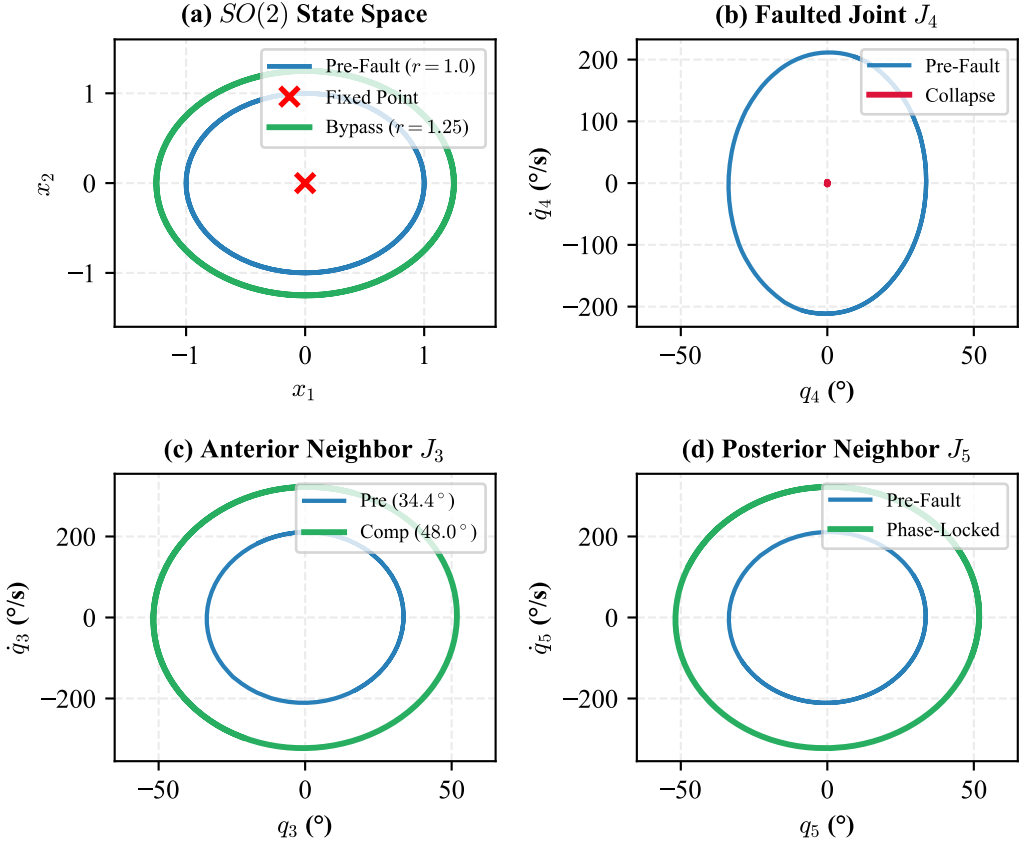

Spinal Limit-Cycle Phase Portraits
(a) $SO(2)$ State Space
Pre-Fault ($r = 1.0$)
Fixed Point
Bypass ($r = 1.25$)
x_1
x_2
(b) Faulted Joint J_4
Pre-Fault
Collapse
q_4 (°)
$\dot{q}_4$ (°/s)
(c) Anterior Neighbor J_3
Pre (34.4°)
Comp (48.0°)
q_3 (°)
$\dot{q}_3$ (°/s)
(d) Posterior Neighbor J_5
Pre-Fault
Phase-Locked
q_5 (°)
$\dot{q}_5$ (°/s)